Gradient Descent and Optimization

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

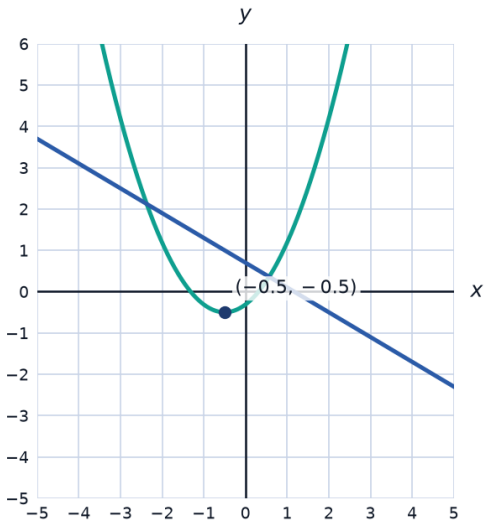
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

2 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

[The Multivariable Chain Rule, Directional Derivatives & Gradient Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



3 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

[Multivariable Optimization: Critical Points, Hessians & Lagrange Multipliers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

4 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Use the method of Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 25$.

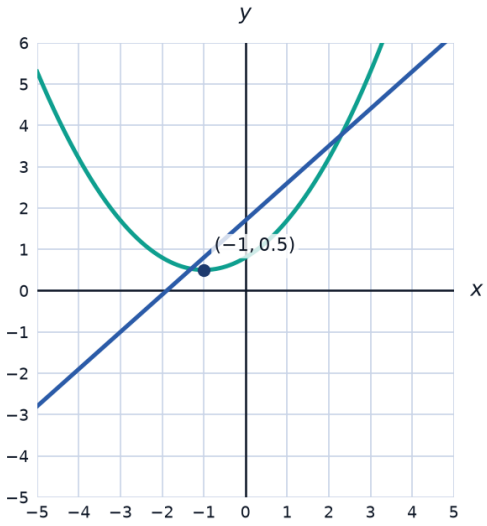
5 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

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6 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Let $z = x^2y - y^3$, where $x(t) = \cos(2t)$ and $y(t) = \sin(2t)$. Use the multivariable chain rule to find $\frac{dz}{dt}$ when $t = 0$.



7 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

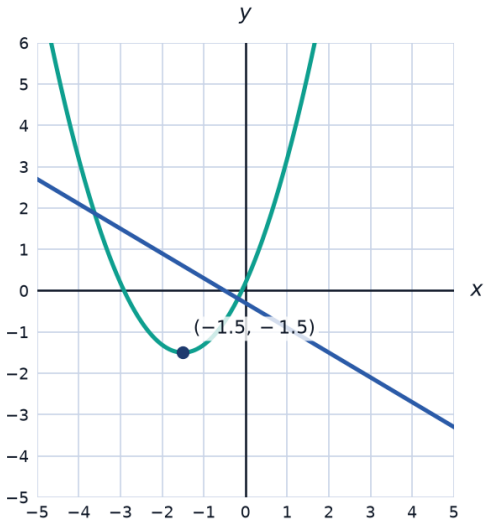
9 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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10 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



11 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

12 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

13 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

14 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

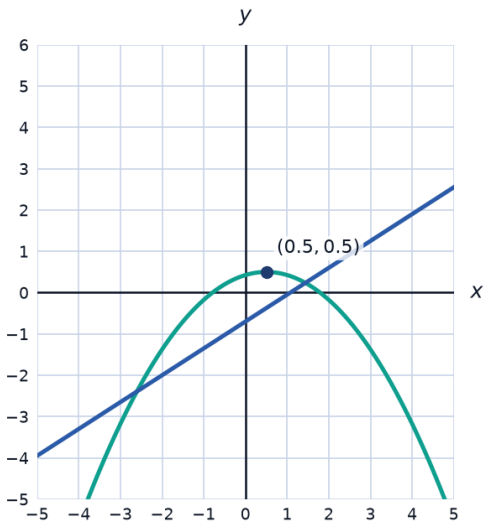
15 **[Challenging]** • **Target: Gradient Descent & Optimization Practice Exercises** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

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16 **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS**

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

18 **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification**

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

[AP Statistics Exam Format & Study Design · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI**

[Unit 1 · Exploring One-Variable & Two-Variable Data · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value \(w = 10 \). If the gradient of the loss is \(\frac{\partial L}{\partial w} = 4 \) and the learning rate is \(\eta = 0.1 \), find the updated weight \(w' \).

a. \(10.40 \)

b. \(9.60 \)

c. \(6.00 \)

d. \(9.90 \)

22 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function \(L(w) = 2w^2 - 8w + 5 \). Find the analytical gradient \(\nabla L(w) \) evaluated at \(w = 2 \).

a. \(\nabla L(2) = 16 \)

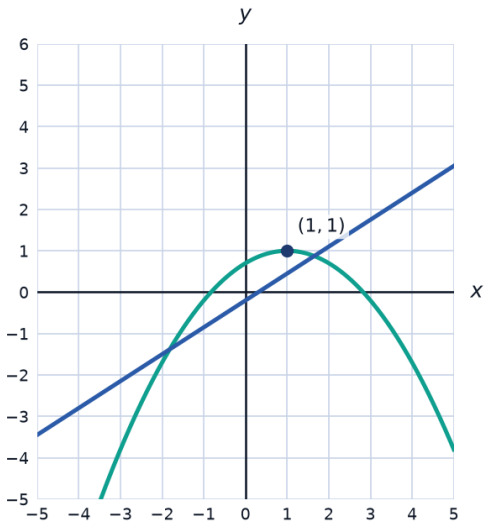
b. \(\nabla L(2) = 8 \)

c. \(\nabla L(2) = 0 \)

d. \(\nabla L(2) = -4 \)

23 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.



a. $ablaL = [10, -8]^T$

b. $ablaL = [8, 6]^T$

c. $ablaL = [4, -3]^T$

d. $ablaL = [8, -6]^T$

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24 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In training a language model token predictor, a binary token occurs $k = 3$ times across $n = 10$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

a. $\hat{p}_{\text{MLE}} = 0.300$

b. $\hat{p}_{\text{MLE}} = 0.450$

c. $\hat{p}_{\text{MLE}} = 0.150$

d. $\hat{p}_{\text{MLE}} = 0.700$

25 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

a. 0.0450

b. 0.9500

c. 0.8500

d. 0.0050

26 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

a. $\hat{\beta}_1 = 0.3333333333333333,\ \hat{\beta}_0 = 10$

b. $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 10$

c. $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$

d. $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 5$

27 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?

a. 75.0%

b. 60.0%

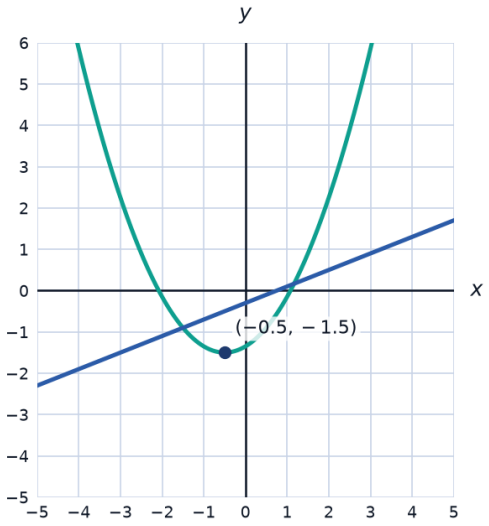
c. 25.0%

d. 85.7%

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28 [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.



a. 81.0%

b. 81.8%

c. 52.4%

d. 13

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is $\mathrm{SSE} = 20$ and the Total Sum of Squares is $\mathrm{SST} = 100$. What is the coefficient of determination R^2?

a. $R^2 = 0.80$

b. $R^2 = 0.20$

c. $R^2 = 0.90$

d. $R^2 = 0.25$

30 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

a. 0.140

b. 0.950

c. 0.279

d. 0.391

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31 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component PC_1?

a. 62.0%

b. 86.0%

c. 24.0%

d. 72.1%

32 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{50}$ onto its top 6 principal components. What is the dimension of each transformed data point?

a. 25

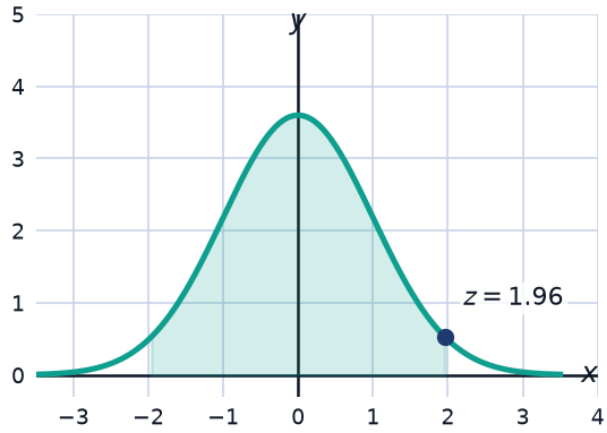
b. 50

c. 44

d. 6

33 **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean μ and variance σ^2 of this feature?



a. $\mu = 25,\ \sigma^2 = 36$

b. $\mu = 36,\ \sigma^2 = 25$

c. $\mu = 25,\ \sigma^2 = 6$

d. $\mu = 6,\ \sigma^2 = 36$

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34 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

a. 0.429

b. 0.353

c. -2.347

d. -0.847

35 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

a. 46

b. 49

c. 42

d. 28

36 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In a dataset, the prior probability of class C_1 is $P(C_1) = 0.10$. What is the probability of the complement class $P(egC_1)$?

a. 0.90

b. 0.95

c. 0.05

d. 1.10

37 [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

a. $(0.5, 0.5)$

b. $(0, 0)$

c. $(-1, -1)$

d. $(1.5, 1.5)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Gradient descent update $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf$ converges to:

a. a local min for smooth f

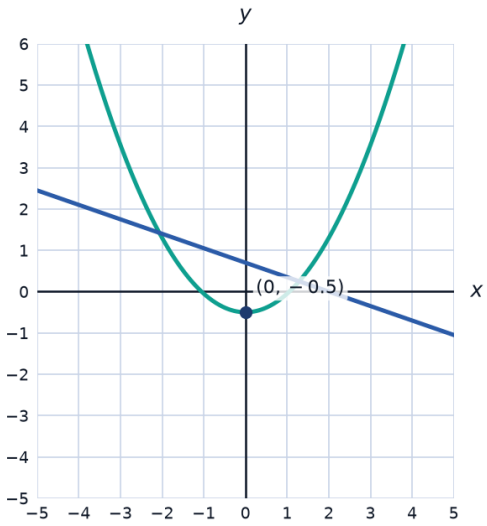
b. global min always

c. a saddle point

d. the origin

39 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.



a. (2, 3)

b. (-2, -3)

c. (4, 6)

d. (0, 0)

40 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.

a. 10

b. 13

c. 26

d. 52

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41 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

a. 1

b. 2

c. 3

d. 4

42 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

a. 2

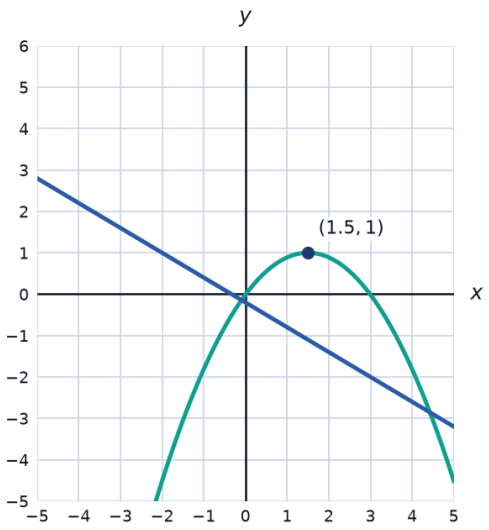
b. 4

c. 8

d. 10

43 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.



a. 2

b. 4

c. 6

d. 0

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44 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of f with respect to y, denoted as $\frac{\partial f}{\partial y}$?

a. $2x - 3y$

b. $2x - 3xy + 4y$

c. $-3x + 4y$

d. $2x - 3y + 4y$

45 [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

a. 6

b. 12

c. -6

d. 15

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